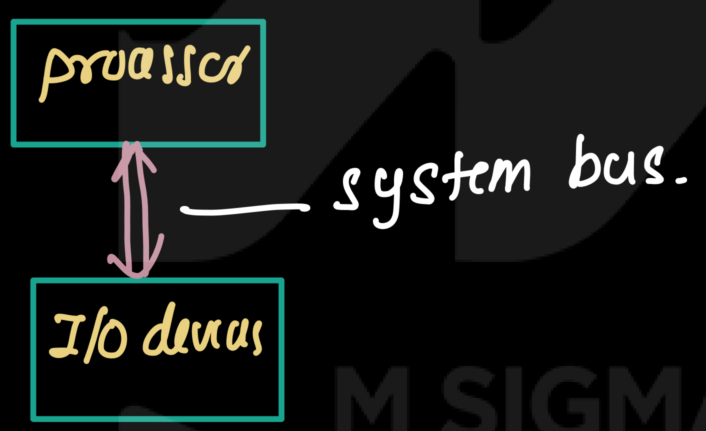
IO structure of Computer System (Note)

- A computer based system mainly consists of 3 components
  1. Processor (CPU)
  2. Memory
  3. Input /output (I/O) devices.

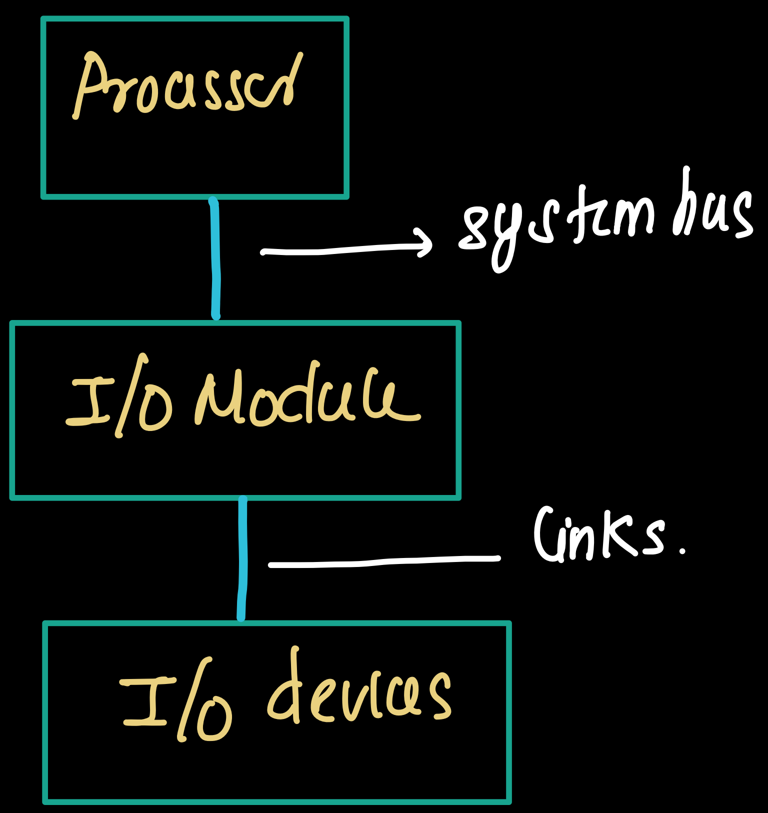
- Processor fetch the instruction from the memory and execute them. During the execution processor need to interract with external devices such as keyboar, mouse.
- To communicate with these device, the processor must access them through an interface
- One possible method is to connect processor and I/O devices directly using system Bus



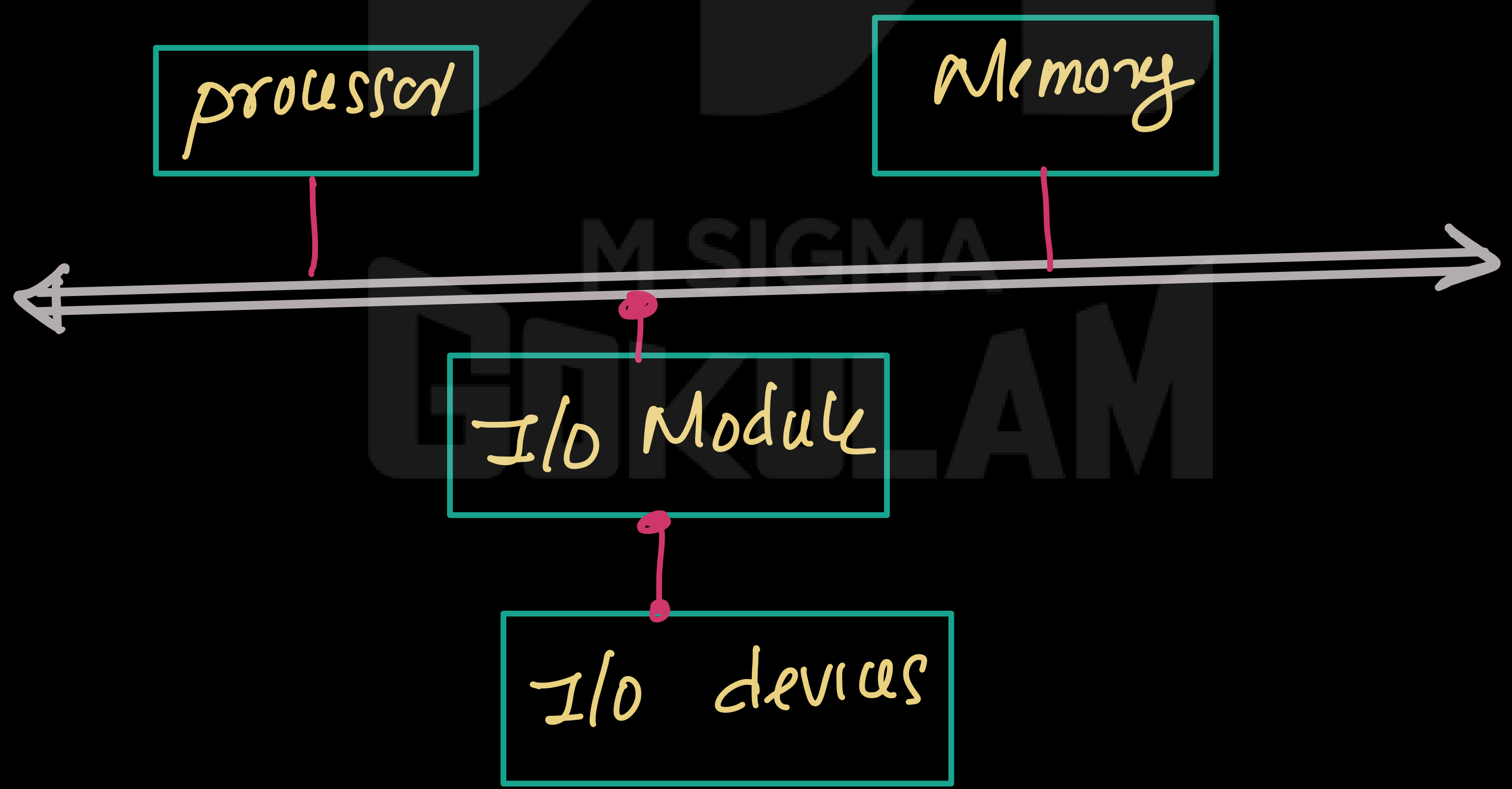
- But direct communication b/w the processor and I/O devices is not efficient because of the following reasons.
  1. Processor operates at very high speed and most of the I/O devices are slow. So there is a speed difference b/w processor and I/O device.
  2. Because of the speed Mismatch, processor would have to wait for the slower devices, which reduce the system efficiency.

Therefore direct connection b/w processor and I/O devices using system bus is not practical.

- To solve this problem, An I/O Module is used. This I/O module act as interface b/w processor and I/O devices



- Processor is connected to the I/O module through the system bus
- The I/O module is connected to the I/O devices through link
- external devices connected to the I/O modules are called peripheral devices.

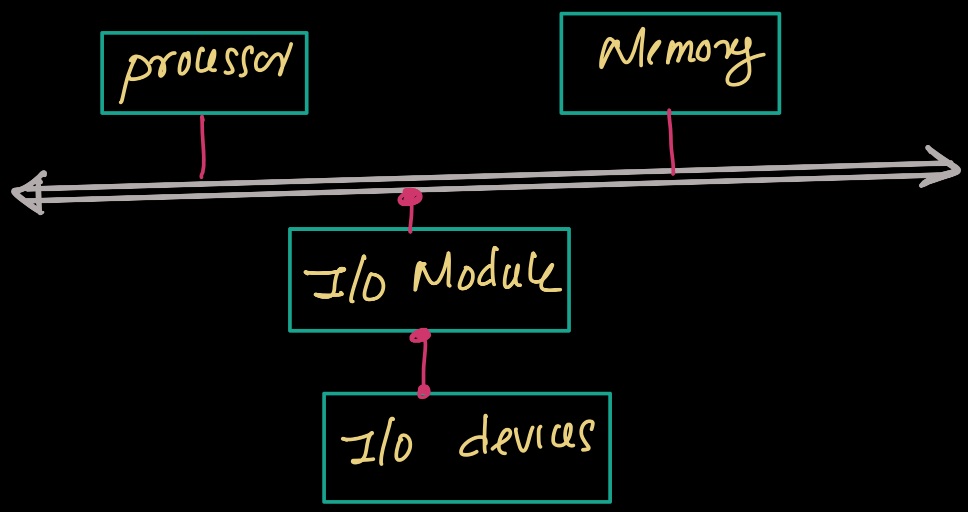


I/O operations

- An I/O operation refers to the ① process of transferring data b/w computer s/m and external devices such as keyboard, printer.
  ② process of transferring data b/w memory and I/O devices

External devices (Note)

Basic computer system consists of processor, memory, s/m bus
I/O Modules, I/O devices.



External devices

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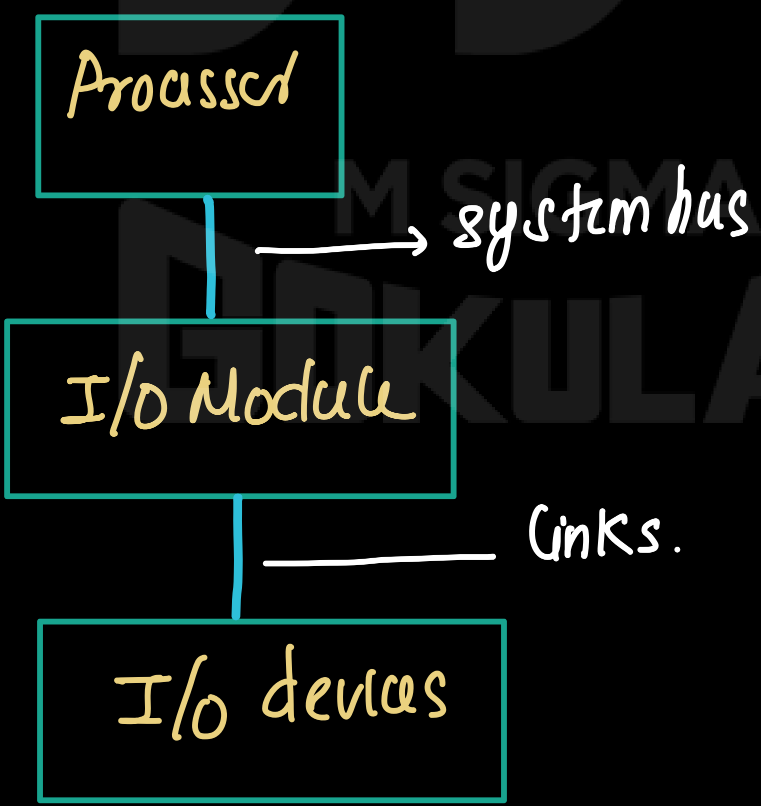
- external devices are the hardware components that allow a computer systems to communicate with the out side environment.
- these devices allow the computer s/m to receive the data, send result and interract with users.
- external devices are not directly connected to the processor. because processor operates at a very high speed while most general devices operates slowly.

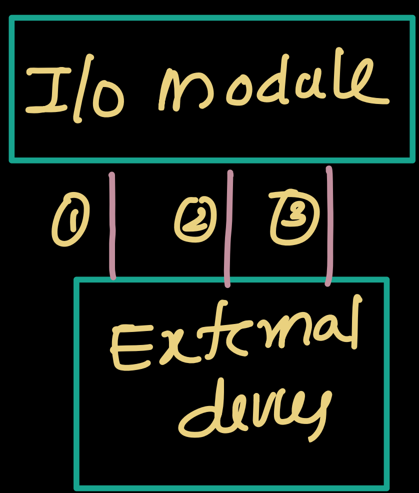
→ Therefore external devices are connected through I/O modules.

- The processor communicates with the I/O module through the system bus and the I/O module communicate with the external devices through a device interface link.



- the connection b/w external devices and I/O modules uses three types of signal

① Control signal
② Data signal
③ Status signal



Structure of External device

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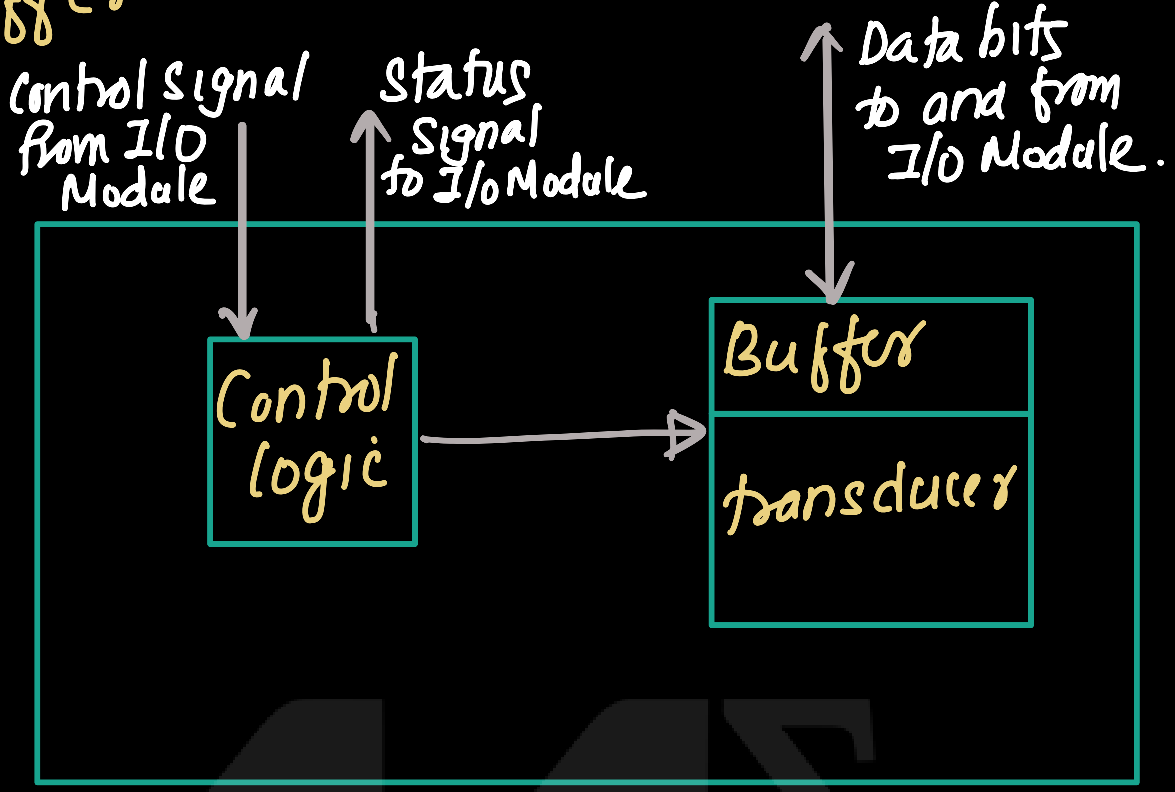
- External device consists of 3 main components
  1. Control logic
  2. transducer
  3. Buffer



Eg: processor needs to read data from I/O device

- The processor send command to the I/O module through the system bus. this command specifies that the processor wants input data from the device
- After receiving command I/O module send a control signal to the I/O devices.
- Control logic inside the external device receive the control signal from the I/O module.
- Control logic control the operation of the device according to the instructions received from the I/O module

→ transducer inside the external device continuously monitor the physical environment and convert this physical phenomena to electrical signal for input operation.

→ Control logic inside the external device continuously monitor the signal from the transducer

→ when control logic detect a valid signal, then control logic convert this signal into bit pattern ⓝ

- This bit patterns are temporarily stored in the buffer attached to the transducer.
- external device send a status signal to the I/O module indicating that data is available
- Then buffered data is transferred to the I/O module through data lines.

① Control logic

- Control logic inside the device manages the operation of the devices
- works according to the instructions from the I/O module.

② Transducer.

Tranducer inside the device convert electrical signal to other form of energy and vice versa

③ Buffer

Buffer temporarily store data during transfer b/w external device and I/O module.

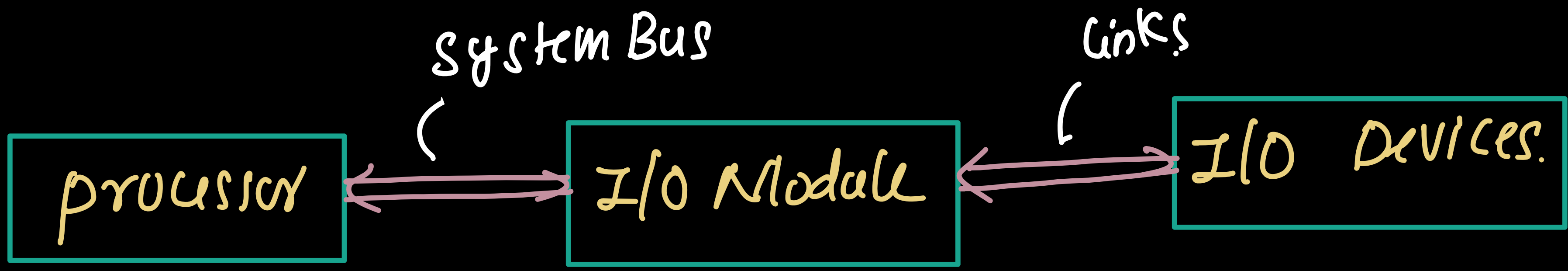
Keyboard input Process

- when key is pressed, it generates an electronic signal
- the transducer in the keyboard convert this signal into electrical signal
- control logic continuously monitor the signal at the transducer. If control signal detect a valid signal. then this signal is translated into bit pattern.
- these bit pattern send to the I/O Module.

I/O Module (Note)

→ I/O module act as interface between processor and external devices such as keyboard, printer etc....

— Since external devices operate much slower than the processor, I/O module manages communication and data transfer b/w the processor and these I/O devices.



— The processor communicates with I/O module through the system bus (Add Bus, data Bus, Control Bus).

→ I/O module communicate with external device through external device interface.

Structure of I/O Module.

— I/O Module consist of following components that manage the communication b/w processor and external devices.

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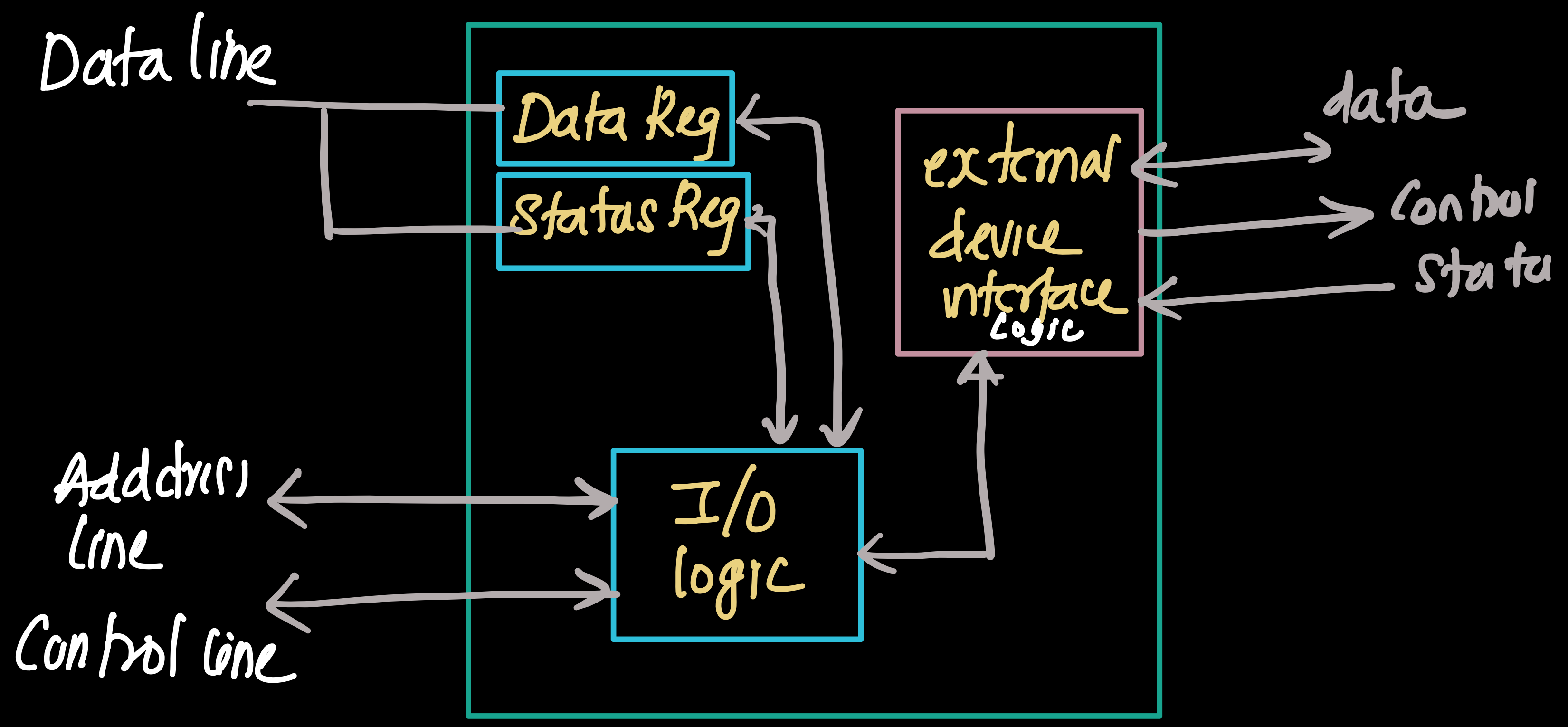


① External device interface logic

- this manages the communication b/w I/O module and external device.
- It handles 3 types of signal
  1. data signal :
  2. Control signal
  3. status signal.

② Data Register

- Data Registers are temporarily store the data that is being transfer between the external device and processor.
- Eg: when key is pressed on keyboard, the generated bit pattern stored in the data register.

② status Register

→ this register store the information about of the device.
→ eg: If key is pressed, then bit pattern stored in the data register, and update the content

of status register (device is redy). N

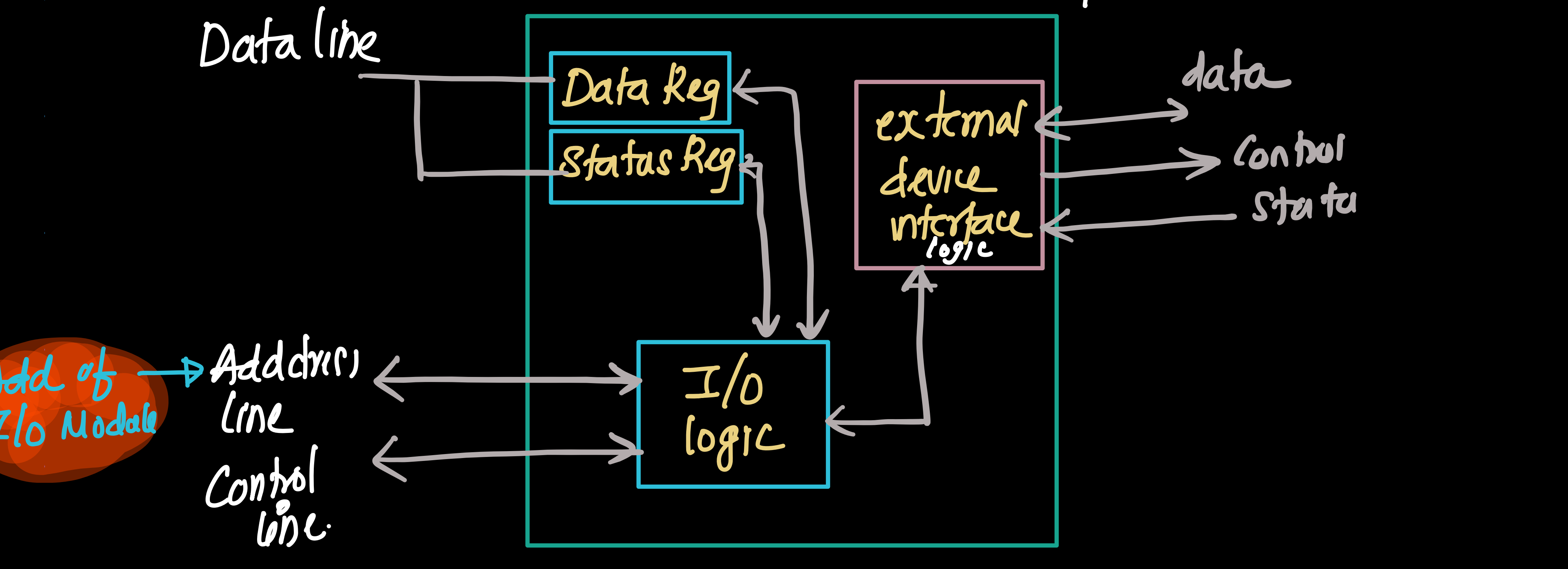
- information includes
  1. ready - device is ready to transfer
  2. Busy - device is busy

3. I/o logic

- I/o logic Control the operation of I/o module.
- → it perform the following functions.
  1. Control the data transfer b/w the processor and external device
  2. Co-ordinate Communication b/w external device interface logic and Registers

Data transfer from I/o device to the processor.

- Consider an I/o device is connected to the processor through I/o module
- Eg: Reading a character from the key board
- processor place the Address of I/o module on the address line of the system bus
- This select specific I/o module and device that processor wants to access



- processor activate control signal on the on the control bus

  eg: To read data from the device, processor activate I/O Read signal

  To send data to the device, processor activate I/O write signal

  these signal specify the type of operation the processor wants the I/O module to perform.

  (processor send activate IO Read signal to the control line of system bus

- I/O logic inside the I/O module decode the control signal. From this it understand that the processor is requesting a read operation.
- After decoding control signal, I/O module sends the appropriate control signal to the external device through the external device interface logic. this tells the device to provide the required data.
- the device sends the data to external device interface logic.
- external device interface logic receives the data and temporarily store in the data Register and update the status register, indicating that input data is available (Ready bit of status register become 1).

  Ready bit = 0 (no data available).

  Ready bit = 1 (data is available)

- After sending read command, processor read the status register of I/O module. Status register contain information such as

- device ready
- device busy.

— processor check whether the device is ready to transmit when the device become ready, I/O module place the data in data lines of system bus and processor read the data

Sample Questions

1. Functions of I/O module
2. structure of I/O Module
3. How data transfer btw an external device and processor